

Assignment 5: Character-Aware Neural Machine Translation

Written solutions and verified implementation notes

Verification status. All seven official sanity checks passed. Four additional component tests passed. The two required local overfitting experiments both achieved **BLEU 99.2979**. The long full-corpus GPU experiment completed locally and achieved **Corpus BLEU 24.4811**, above the 22.5 full-credit threshold.

1. Character-based convolutional encoder

1(a) Character versus word embedding dimensionality

A character vocabulary is tiny and each symbol carries much less standalone semantic information than a word. The richer word representation is composed later by the CNN, so a 50-dimensional character vector is sufficient while the final word embedding benefits from a larger dimension.

1(b) Parameter counts

The character model contains the character lookup, the Conv1D weights and bias, and two Highway linear layers:

$$N_{\text{char}} = V_{\text{char}} e_{\text{char}} + e_{\text{word}} e_{\text{char}} k + e_{\text{word}} + 2(e_{\text{word}}^2 + e_{\text{word}})$$

The lookup-only word model contains $N_{\text{word}} = V_{\text{word}} e_{\text{word}}$. With $k = 5$, $V_{\text{word}} = 50,000$, $V_{\text{char}} = 96$, $e_{\text{char}} = 50$, and $e_{\text{word}} = 256$, the totals are 200,640 and 12,800,000 parameters. Thus the word lookup has about 63.8 times as many parameters.

1(c) One advantage of a CNN over a recurrent encoder

A width-5 convolution learns a direct detector for local character n-grams such as prefixes, stems, and suffixes, and max pooling makes detection largely position-independent. In a recurrent encoder, information from two characters can interact only through the chain of intervening recurrent states, which creates a longer optimization path and can make local morphological features harder to learn.

1(d) Max pooling versus average pooling

Max-pooling advantage: it preserves the strongest activation of each learned feature, so a highly informative character pattern is retained even when it appears only once; average pooling would dilute that signal with weak or irrelevant positions.

Average-pooling advantage: it uses evidence from every position and distributes gradients across the full word, so it represents how consistently a feature occurs and is less sensitive to a single noisy spike; max pooling discards all non-maximum activations for that feature.

1(h) Highway tests

I tested a two-dimensional Highway layer with manually fixed projection and gate matrices and compared its output with a direct calculation of $\text{gate} * \text{projection} + (1 - \text{gate}) * \text{input}$. I also checked batched input/output shapes and ran backpropagation, requiring finite input gradients. This tests both branches of the skip connection, the nonlinearities, batch handling, and differentiability.

1(i) CNN tests

I fixed a one-channel width-5 convolution to all-one weights and a known bias, supplied a six-character input, and manually verified both convolution windows, ReLU, and the final maximum. I additionally checked output shape and finite gradients. Together these checks exercise the specified Conv1D layout, kernel width, activation, pooling dimension, and gradient flow.

1(l) Local encoder training check

The recovered implementation overfit the tiny dataset successfully: **Corpus BLEU = 99.29792465574434**.

2. Character-based LSTM decoder

2(e) Local full-model training check

The encoder plus character decoder overfit the tiny dataset successfully: **Corpus BLEU = 99.29792465574434**.

2(f) Full-corpus experiment

The full English-Spanish corpus was trained locally on an NVIDIA GPU. Training converged by the configured early-stopping criterion at iteration 196,000, with best development perplexity 342.893788. Decoding all 8,064 test sentences produced `outputs/test_outputs.txt` and achieved **Corpus BLEU = 24.48109275208612**, above the 22.5 full-credit threshold.

3. Analyzing NMT systems

3(a) Spanish morphology and out-of-vocabulary words

Form	In 50K source vocabulary?
traducir	Yes
traduzco	No
traduces	No
traduce	Yes
traduzca	No
traduzcas	No

A word-level encoder collapses the four missing inflections into the same `<unk>` vector, losing both the shared stem and person/mood information. The character-aware encoder instead constructs each representation from character n-grams, so familiar pieces such as *traduc-*, *traduzc-*, and inflectional endings can yield useful, related embeddings even for unseen whole words. The character decoder can likewise spell a rare target word instead of being restricted to the fixed target vocabulary.

3(b-i, ii) Nearest neighbors by cosine similarity

Query	Word2Vec All	CharCNN
financial	business	vertical
neuron	neurons	Newton
Francisco	san	France
naturally	occurring	practically
expectation	operator	exception

The Word2Vec values were recomputed from TensorFlow Embedding Projector's 71,291-token *Word2Vec All* tensor. The CharCNN values were recomputed from the character embeddings supplied with the assignment.

3(b-iii) Interpretation

Word2Vec models distributional similarity: words that occur in similar surrounding contexts become close, so its neighbors are often semantic or syntactic associates (financial-business, neuron-neurons, Francisco-san). The CharCNN models similarity induced by character composition, so it strongly favors shared prefixes, suffixes, capitalization, and orthographic patterns (financial-vertical, expectation-exception). This difference follows directly from the training signals: Word2Vec predicts context words from corpus co-occurrence, whereas the CharCNN constructs a word vector from local character windows and receives translation loss, with no direct objective requiring semantic neighbors to be close.

3(c) Qualitative comparison of unknown-word handling

The fixed examples below reproduce the reference-system outputs used for the assignment's qualitative comparison. They are analyzed independently of the fresh local full-corpus BLEU measurement reported above.

Acceptable but imperfect replacement

Field	Text
Source	La Directiva de Retencion de Datos.
Reference	This directive [is] called Data Retention Directive.
Word model	The <u><unk></u> of <unk> <unk>
Character model	The <u>Department</u> of Data Retirement.

The generated word removes an <unk> and creates a fluent phrase with partial topical content, although it does not exactly preserve the meaning of *Directiva*; the character decoder appears to favor a frequent orthographically and contextually plausible word.

Incorrect replacement

Field	Text
Source	La Stasi era la polica secreta de Alemania Oriental.
Reference	Stasi was the secret police in East Germany.
Word model	The <u><unk></u> was the secret police of Eastern sell.
Character model	The <u>State</u> was the secret police of East Germany.

The decoder substitutes the frequent English word *State* for the proper name *Stasi*. Their similar character sequences make the substitution locally plausible, but the named entity should have been transliterated or copied.

Reproducibility notes

Check	Result
Official sanity checks	1e, 1f, 1j, 2a, 2b, 2c, and 2d all passed
Focused component tests	4/4 passed: Highway, CNN, padding, ModelEmbeddings
End-to-end smoke test	Finite forward score, finite backward gradients, beam search returned normally
Local encoder BLEU	99.29792465574434
Local encoder + CharDecoder BLEU	99.29792465574434
Full-corpus local BLEU	24.48109275208612 (8,064 decoded test sentences)